DIGITAL FORENSICS · IMAGE EXAMINATION

Image Forensic Examination Report

Evidence file: **spliced_demo.jpg**

CASE ID	EXAMINER	GENERATED (UTC)	TOOL
PP-20261103-175325-8AA0	Not recorded	2026-11-03 17:53:25	PixelProof 1.0.0

Likely tampered

A strong forensic signal is consistent with tampering; corroborate before concluding.

Confidence 74% · coverage 3 module(s) · agreement 16%

54.8
out of 100

Banding: score 50 or above · final score blends the weighted consensus (30.7) with the strongest single module (84.3).

1. Executive summary

PixelProof executed **3** analysis module(s) in **0.4 s**. The combined manipulation score is **54.8/100**, placing the image in the **Likely tampered** band (score 50 or above). The strongest contribution came from **Error Level Analysis** (84.3/100). **3** finding(s) above info level were recorded: 1 high, 0 medium, 2 low.

Signals supporting manipulation

- Error Level Analysis scored 84.3/100 - mean residual 0.30 · p99.5 4.0 · 1 dense region(s) · outlier fraction 16.1%

Signals against manipulation

- File analysis & integrity scored 0.0/100 - JPEG · 1200×800 · 54.5 KB · signature/extension match
- Metadata / EXIF scored 0.0/100 - 22 EXIF tags · Canon EOS 90D · GPS 12.9750, 77.5944

Not available in this build: Copy-move detection, Steganography checks. These modules are excluded from the score rather than scored as zero.

2. Evidence and integrity

FILE NAME	spliced_demo.jpg
FULL PATH	`samples/tampered/spliced_demo.jpg`
SIZE	54.5 KB (55,768 bytes)
CONTAINER	JPEG · magic JPEG · extension .jpg (match)
DIMENSIONS	1200 x 800 px · 0.96 MP · RGB mode
FRAMES	1
FILESYSTEM MODIFIED	2026-11-03T12:12:49.049504+00:00
ANALYSIS WINDOW	2026-11-03T17:53:24.645145+00:00 to 2026-11-03T17:53:25.089602+00:00 (0.44 s)
SHA-256	`8aa0649f142b73810bbc498c00b5dbd2a96c4f304e54aae5a07d0d4753286dde`
SHA-1	`c31e5d35200568b1fb7be940f2febf84d8e048ef`
MD5	`acbb56f51ad2214fd6e3d4a2624c79c1`

Hashes cover the exact bytes on disk. Re-hashing the same file later must reproduce SHA-256 exactly; any difference invalidates this report's link to the evidence.

3. Module results

Module	Status	Weight	Signal	Score	Summary
File analysis & integrity	OK	0.27		0.0	JPEG · 1200×800 · 54.5 KB · signature/extension match
Metadata / EXIF	OK	0.36		0.0	22 EXIF tags · Canon EOS 90D · GPS 12.9750, 77.5944
Error Level Analysis	OK	0.36		84.3	mean residual 0.30 · p99.5 4.0 · 1 dense region(s) · outlier fraction 16.1%

Weights are re-normalised over the modules that ran, so a partial pipeline still produces a meaningful score. The Signal column is each module's own 0-100 manipulation signal, not the final verdict.

4. Findings

HIGH

Localised compression inconsistency

1 dense residual region(s) found. The largest covers 18x14 blocks (288x224 px) around (144, 128)-(432, 352) with mean residual 1.04 versus 0.14 in the surrounding ring (5.5x). This is consistent with a locally edited, pasted or re-encoded region. Confirm with copy-move and noise analysis: texture, sharpening and earlier JPEG saves can produce a similar pattern.

Source: Error Level Analysis · Confidence 91% · ela, tamper, compression

LOW

GPS coordinates embedded

12.975000° N, 77.594444° E · altitude 920.0 m · map https://www.openstreetmap.org/?mlat=12.975000&mlon=77.594444#map=16/12.975000/77.594444. Location is sensitive personal data and a strong corroboration point.

Source: Metadata / EXIF · Confidence 99% · privacy, gps

LOW

File modified well after capture

Filesystem mtime 2026-11-03 12:12:49.049504 is more than two days from DateTimeOriginal 2026-09-10 16:42:11. Copying, exporting and editing all change mtime; compare with the other timestamps.

Source: Metadata / EXIF · Confidence 45% · timestamp

INFO

Cryptographic digests computed

SHA-256 8aa0649f142b73810bbc498c00b5dbd2a96c4f304e54aae5a07d0d4753286dde · SHA-1 c31e5d35200568b1fb7be940f2febf84d8e048ef · MD5 acbb56f51ad2214fd6e3d4a2624c79c1

Source: File analysis & integrity · Confidence 99% · integrity, custody

INFO

Filename and signature are consistent

JPEG content, 1200×800, extension '.jpg' accepted.

Source: File analysis & integrity · Confidence 95% · container

INFO

Camera identity present

Canon EOS 90D · lens EF-S 18-135mm f/3.5-5.6 IS USM · software Canon EOS 90D Firmware 1.0.0

Source: Metadata / EXIF · Confidence 90% · provenance, metadata

5. Visual evidence

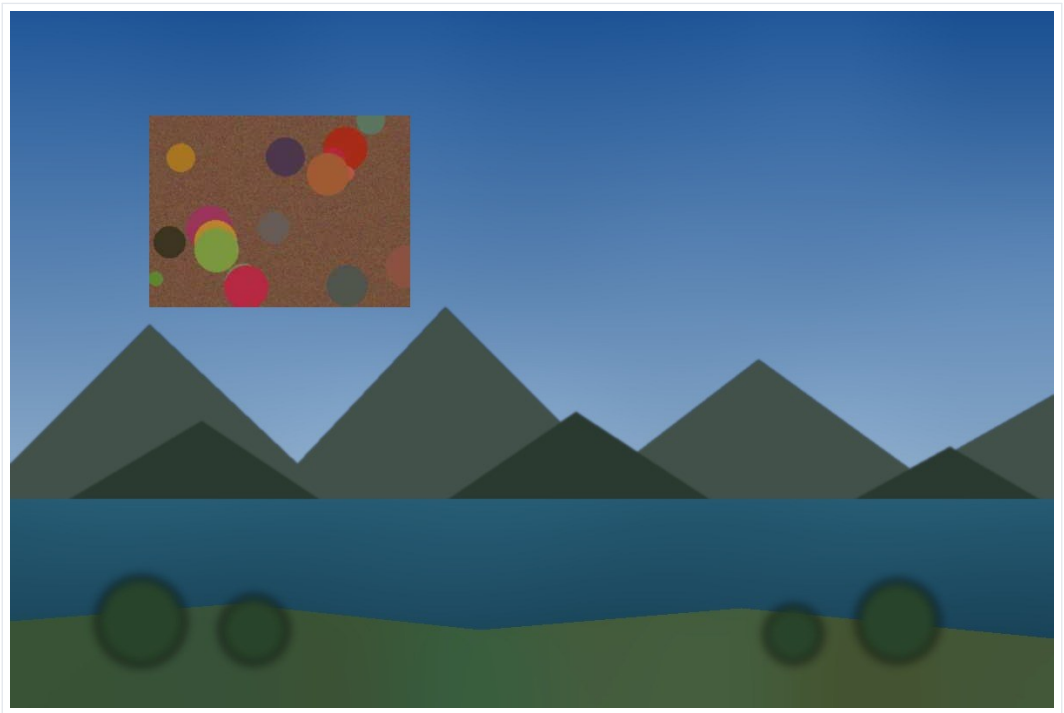


Figure 1 — Evidence image as submitted (display copy only).

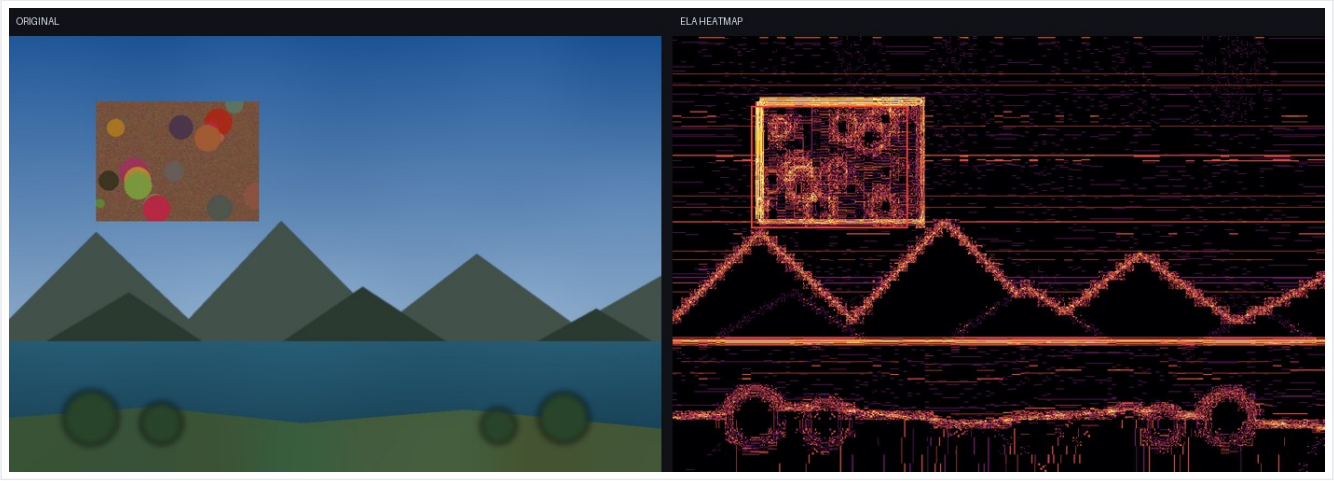


Figure 2 — Original (left) and ELA heatmap (right). Bright areas changed most after recompression at the configured quality.

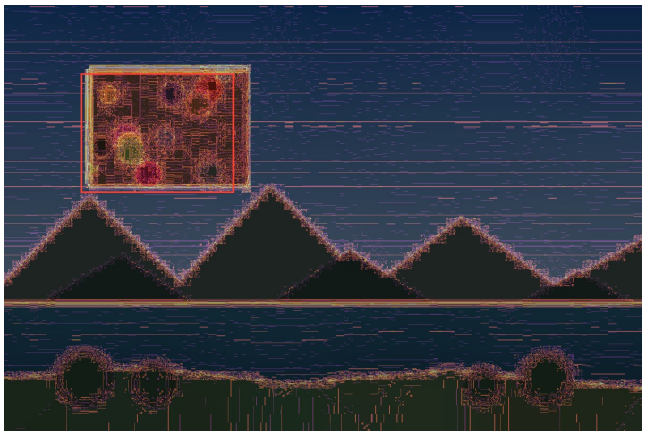


Figure 3 — ELA overlay with detected regions outlined

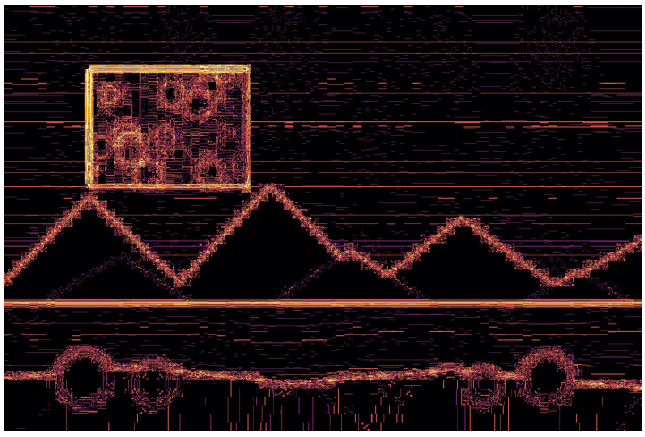


Figure 4 — Colourised ELA residual

6. Metadata detail

Camera and software

MAKE	Canon
MODEL	EOS 90D
LENS	EF-S 18-135mm f/3.5-5.6 IS USM
BODY SERIAL	SN-DEMO-000123
SOFTWARE	Canon EOS 90D Firmware 1.0.0
ARTIST	Sagar
COPYRIGHT	CC BY 4.0 sample

Location

LATITUDE	12.975000 (12°58'30.0"N)
LONGITUDE	77.594444 (77°35'40.0"E)
ALTITUDE	920.0 m
GPS UTC	2026:09:10 10:42:11Z

https://www.openstreetmap.org/?mlat=12.975000&mlon=77.594444#map=16/12.975000/77.594444
Coordinates are shown as stored. The tool does not reverse-geocode online, so the location name is not inferred and no evidence leaves the machine.

Timestamp coherence

Capture time	2026-09-10 16:42:11 UTC
Digitised time	2026-09-10 16:42:15 UTC
Tag change time	2026-09-10 16:42:11 UTC
File modified	2026-11-03 12:12:49 UTC

Provenance keywords

Editing software	none detected
AI generators	none detected
EXIF tags read	22
ICC profile	absent

EXIF tags

Tag	Value
EXIF BodySerialNumber	SN-DEMO-000123
EXIF DateTimeDigitized	2026:09:10 16:42:15
EXIF DateTimeOriginal	2026:09:10 16:42:11
EXIF ExifImageLength	800
EXIF ExifImageWidth	1200
EXIF LensModel	EF-S 18-135mm f/3.5-5.6 IS USM
GPS GPSAltitude	920
GPS GPSAltitudeRef	0
GPS GPSDate	2026:09:10

Tag	Value
GPS GPSLatitude	[12, 58, 30]
GPS GPSLatitudeRef	N
GPS GPSLongitude	[77, 35, 40]
GPS GPSLongitudeRef	E
GPS GPSTimeStamp	[10, 42, 11]
Image Artist	Sagar
Image Copyright	CC BY 4.0 sample
Image DateTime	2026:09:10 16:42:11
Image ExifOffset	198
Image GPSInfo	364
Image Make	Canon
Image Model	EOS 90D
Image Software	Canon EOS 90D Firmware 1.0.0

7. Method and thresholds

Every threshold used below is stored in `utils/config.py`. They are empirical values chosen for this project's sample set, not trained on a labelled corpus. The report records them so another examiner can reproduce the run.

Module	Settings used
File analysis	MD5, SHA-1 and SHA-256 over the exact bytes; magic-byte container detection; extension agreement; double-extension regex; 50 MB and 80,000,000 pixel limits.
Metadata	ExifRead tag extraction; GPS rationals to decimal, DMS and OSM link; timestamp coherence checks; editing and AI keyword lists; EXIF thumbnail dHash compared with the main image (Hamming distance > 16 is a mismatch).
ELA	JPEG re-save at quality 90; 16 px blocks; robust threshold median + 4×MAD with an absolute floor of 0.5; 9×9 density window requiring 50% high-residual blocks; at least 12 blocks; mean residual ≥ 0.7; region/ring contrast ≥ 2.0.
Not available	Copy-move detection, Steganography checks — excluded from scoring in this build.

Verdict formula

```
weighted = sum(weight × module score) / sum(weight)
final = 0.55 × weighted + 0.45 × strongest module
```

Weights: file 0.15, metadata 0.20, ELA 0.20, copy-move 0.30, stego 0.15.

Band	Score	Meaning
Likely authentic	score below 20	No strong tampering signal; still requires examiner judgement.
Suspicious	score 20 to 49	At least one signal warrants review and corroboration.
Likely tampered	score 50 or above	A strong signal is consistent with manipulation.
Inconclusive	not scored	Not enough usable evidence to score.

Limitations

- ELA detects compression-history discontinuity, not intent. Smooth brightness changes and blurring can produce a low score even when the image was edited.
- A negative result is not proof of authenticity. Absence of evidence is not evidence of absence.
- Copy-move and steganography modules are not part of this build, so duplicated regions and LSB payloads are not assessed here.

- EXIF can be forged or stripped. Editor tags are leads, not findings of malicious intent.
- Thresholds are empirical and were not validated against a large labelled corpus.
- This tool is an educational prototype, not court-validated forensic software.

8. Declaration and chain of custody

The evidence file was opened read-only for this examination. All derived images, heatmaps and thumbnails were written under `output/`; the original bytes were never modified. The SHA-256 digest printed in Section 2 can be recomputed at any time to prove that the examined file is unchanged.

EXAMINED BY	Not recorded
DATE (UTC)	2026-11-03 17:53:25
CASE ID	PP-20261103-175325-8AA0

SIGNATURE

Disclaimer

This report is the output of an automated screening tool. Findings are investigative leads, not legal proof. A qualified examiner must interpret them alongside the original evidence, the chain of custody and any corroborating sources. Scores are relative to the documented thresholds and do not express a probability of tampering.

Reproduction

```
python -m forensics.report samples/tampered/spliced_demo.jpg
python -m unittest discover -s tests -v
```

— End of report —